

Operational Research: Tutorial 3

Linear Programming 2

Exercises

1) Solve the LP using the Problem-Based approach in Pyomo

$$\begin{aligned} &\text{maximise} && 12x + 18y \\ &\text{subject to:} && \\ &&& 2x + 4y \leq 55 \\ &&& 2x + y \leq 25 \\ &&& x, y \geq 0 \end{aligned}$$

a. Let x and y be restricted to integer values, solve the Integer Linear Program (ILP) using Pyomo

```
Variables:
  x : Size=2, Index={0, 1}
      Key : Lower : Value : Upper : Fixed : Stale : Domain
        0 :      0 :   7.0 :   None : False : False : NonNegativeIntegers
        1 :      0 :  10.0 :   None : False : False : NonNegativeIntegers

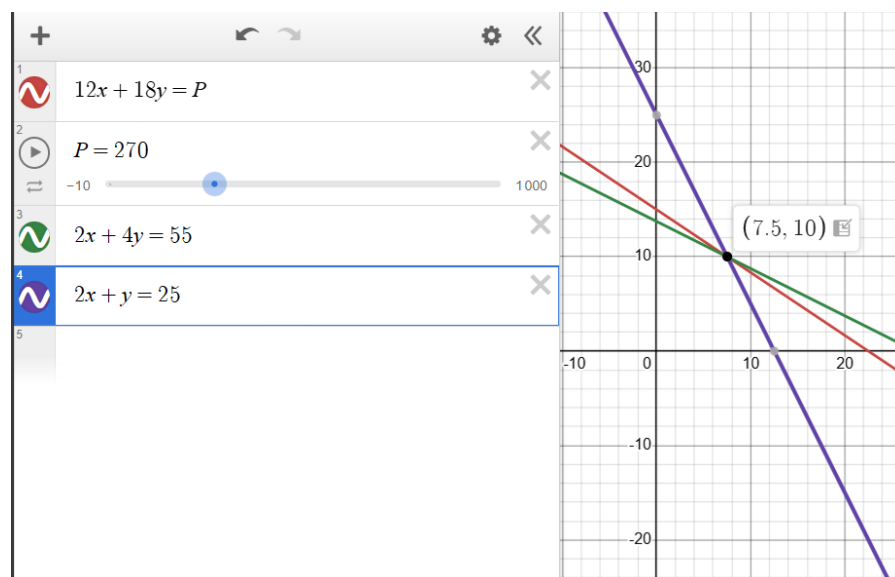
Objectives:
  obj : Size=1, Index=None, Active=True
      Key : Active : Value
      None :   True : 264.0
```

b. Let x be a real value and y restricted to an integer, solve the Mixed-Integer Linear Program (MILP) using Pyomo

```
Variables:
  x : Size=1, Index=None
      Key : Lower : Value : Upper : Fixed : Stale : Domain
      None :      0 :   7.5 :   None : False : False : NonNegativeReals
  y : Size=1, Index=None
      Key : Lower : Value : Upper : Fixed : Stale : Domain
      None :      0 :  10.0 :   None : False : False : NonNegativeIntegers

Objectives:
  obj : Size=1, Index=None, Active=True
      Key : Active : Value
      None :   True : 270.0
```

Note: You can graphically plot this to validate your solution.



- 2) C&S (Management Consultants) have been employed to help the *Icki-Toffee* Company to develop a popular, economical but more health-conscious selection of candies to add to their scrumptious range of toffees. There has been much publicity that the Government are so worried about obesity that they plan to introduce a coding system for all food products which will inform the public about products containing foods which may be adding to the problem. There are rumours that the Government aim to strongly promote the concept of fruit as a healthy alternative to sweets, so *Icki-Toffee* are very keen to introduce fruit, in the form of fillings, soft centres etc into their range. They are determined to ensure that at least one of their selections of candies does not fall into the worst category.

The selection chosen must weigh more than 320g and satisfy certain basic restrictions on fat and calorific content. These are outlined below:

- Total fat content by weight must be less than 45% of the total weight of the selection.
- Calorific value of the selection must be less than 2850 calories.

The quantities of fat and calories in each candy are given in Table 1.

The costs of each candy are given in Table 2.

Table 2 also lists the results of an in-store survey which Icki-Toffee carried out in several of its retail outlets. The popularity number is measured on the scale where 10 is very popular and 0 not popular.

- a) **Formulate a model (using summation notation) given that the aim is to minimise the cost of the selection. Is this a good selection? ***

minimise

$$\sum_i C_i x_i$$

subject to:

$$\sum_i W_i x_i \geq 320$$

$$\sum_i K_i x_i \leq 2850$$

$$\sum_i F_i x_i \leq 0.45 \sum_i W_i x_i$$

and

$$\begin{aligned} x_i &\geq 0, & \forall i \\ x_i &\in \mathbb{Z}, & \forall i \end{aligned}$$

b) Obtain the optimal selection via Pyomo

Variables:

```
x : Size=12, Index={0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}
  Key : Lower : Value : Upper : Fixed : Stale : Domain
    0 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    1 :      0 :   9.0 :   None : False : False : NonNegativeIntegers
    2 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    3 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    4 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    5 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    6 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    7 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    8 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    9 :      0 :  10.0 :   None : False : False : NonNegativeIntegers
   10 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
   11 :      0 :   3.0 :   None : False : False : NonNegativeIntegers
```

Objectives:

```
obj : Size=1, Index=None, Active=True
  Key : Active : Value
  None :      True : 268.0
```

c) Obtain the optimal selection given that the aim is to maximise the popularity of the selection via Pyomo, what is the weight of the selection? **

Variables:

```
x : Size=12, Index={0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}
  Key : Lower : Value : Upper : Fixed : Stale : Domain
    0 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    1 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    2 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    3 :      0 :   5.0 :   None : False : False : NonNegativeIntegers
    4 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    5 :      0 :  50.0 :   None : False : False : NonNegativeIntegers
    6 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    7 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    8 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
    9 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
   10 :      0 :   0.0 :   None : False : False : NonNegativeIntegers
   11 :      0 :   5.0 :   None : False : False : NonNegativeIntegers
```

Objectives:

```
obj : Size=1, Index=None, Active=True
  Key : Active : Value
  None :      True : 365.0
```

We can now calculate the weight of our selection, above we selected 5 of Coconut Ice, 50 of Lime Sherbet and 5 Grape Bunch.

$$5 \times 12 + 50 \times 10 + 5 \times 15 = 60 + 500 + 75 = 635$$

Alternatively we can look at the constraints from the output and see that it is 635:

```

Constraints:
weight_min : Size=1
  Key : Lower : Body : Upper
  None : 320.0 : 635.0 : None
cals_max : Size=1
  Key : Lower : Body : Upper
  None : None : 2850.0 : 2850.0
fat_weight_min : Size=1
  Key : Lower : Body : Upper
  None : None : -0.75 : 0.0

```

d) Comment on whether these are good selections. Would you be happy?

Have a think about this yourself, this isn't for me to say, but I will comment on it in the next lecture. Is there a flaw in our modelling, what about our objective, is it really maximising popularity?

Candy - x_i	Fat (g) - F_i	Calories - K_i	Weight (g) - W_i
Buttery butterscotch	20	100	24
Caramel softie	16	100	22
Orange dream	10	80	15
Coconut Ice	7	50	12
Lemon sorbet	6	60	10
Lime sherbet	5	40	10
Strawberry surprise	13	80	14
Raspberry mousse	15	90	17
Ginger toffee	18	80	20
Rose-water posy	-	145	8
Walnut whirl	19	120	20
Grape bunch	-	120	15

Table 1: Table to show content information about each candy

Candy - x_i	Cost - C_i	Popularity - P_i
Buttery butterscotch	12	10
Caramel softie	12	10
Orange dream	12	8
Coconut Ice	15	9
Lemon sorbet	13	5
Lime sherbet	13	6
Strawberry surprise	16	7
Raspberry mousse	16	8
Ginger toffee	15	10
Rose-water posy	10	3
Walnut whirl	18	8
Grape bunch	20	4

Table 2: Cost and popularity data.